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A LAB REPORT

Genetic Algorithm for a Painter Robot

Artificial Intelligence - Lab Sheet No. 6

SUBMITTED BY

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Submitted to: Department of Electronics and Computer Engineering

This report contains the implementation, plots, and analysis requested in Assignments
1-4.

1. Objectives

The objective is to use a genetic algorithm to evolve a local rule set for a painter robot. The robot receives a chromosome of 54 actions, observes the current, forward, left, and right cells, and tries to cover as much paintable floor as possible without repeatedly visiting the same cells.

2. Assignment 1: Hand-designed strategy

A strong strategy for an empty rectangular room is a serpentine or lawn-mower sweep. The robot starts near a corner, moves straight across one row, turns at the boundary, shifts by one row, and traverses the next row in the opposite direction. Repeating this pattern visits every floor cell once apart from the turns and therefore gives nearly complete coverage with very little wasted movement. This is a conceptual strategy only; it was not manually encoded into the chromosome.

3. Representation and simulation

The state is encoded as $[c, f, l, r]$. The current cell c has two values, 0 for unpainted and 2 for painted; the other cells each have three values, 0 for empty, 1 for wall or furniture, and 2 for painted. Consequently, the chromosome has $2 \times 3 \times 3 \times 3 = 54$ genes. Each gene stores one of four actions: 0 = no turn, 1 = turn left, 2 = turn right, and 3 = choose left or right randomly. The painter observes the state, applies its turn, paints the current unpainted cell, and attempts to move forward. Outside the room is treated as a wall.

```
state_index = (((current_index * 3 + forward) * 3 + left) * 3 + right)
fitness(chromosome) = mean(covered_cells / paintable_cells) over 3 fixed starts
```

The room is 20 x 40. The empty-room run contains 800 paintable cells. The furnished room uses four rectangular pieces of furniture with a total area of 100 cells, leaving 700 paintable cells. Each chromosome is evaluated from three fixed random starts during training and eight independent validation starts. A three-pass step limit prevents a looping policy from running forever.

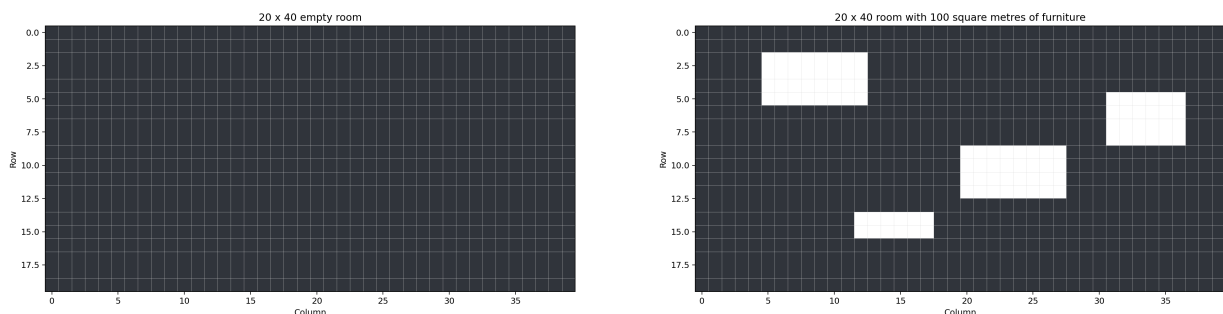


Figure 1: Empty room and furnished room used in the experiments.

4. Genetic algorithm

The initial population contains 50 chromosomes generated uniformly from actions 0-3. For each generation, chromosomes are ranked by average efficiency. Two elite chromosomes are copied unchanged. Parents are selected by a size-3 tournament, and each pair produces children using a single random crossover point. Every locus then mutates independently with probability 0.002. This process is repeated for 200 offspring generations.

Parameter	Value
Population	50
Chromosome length	54
Generations	200
Training episodes	3
Mutation rate	0.002 per locus
Selection	Tournament, size 3, with 2 elite copies
Crossover	Single-point
Random seed	20260802

5. Assignment 2: Empty-room experiment

The empty-room population learns a high-coverage policy quickly. The average and median curves rise sharply in the first few generations, which indicates that crossover and selection discover useful turning patterns early. The best policy reaches 100% on at least one validation start.

Measure	Empty-room result
Paintable cells	800
Mean validation efficiency	99.08%
Best validation efficiency	100.00%
Worst validation efficiency	94.12%
Mean steps	2286.75

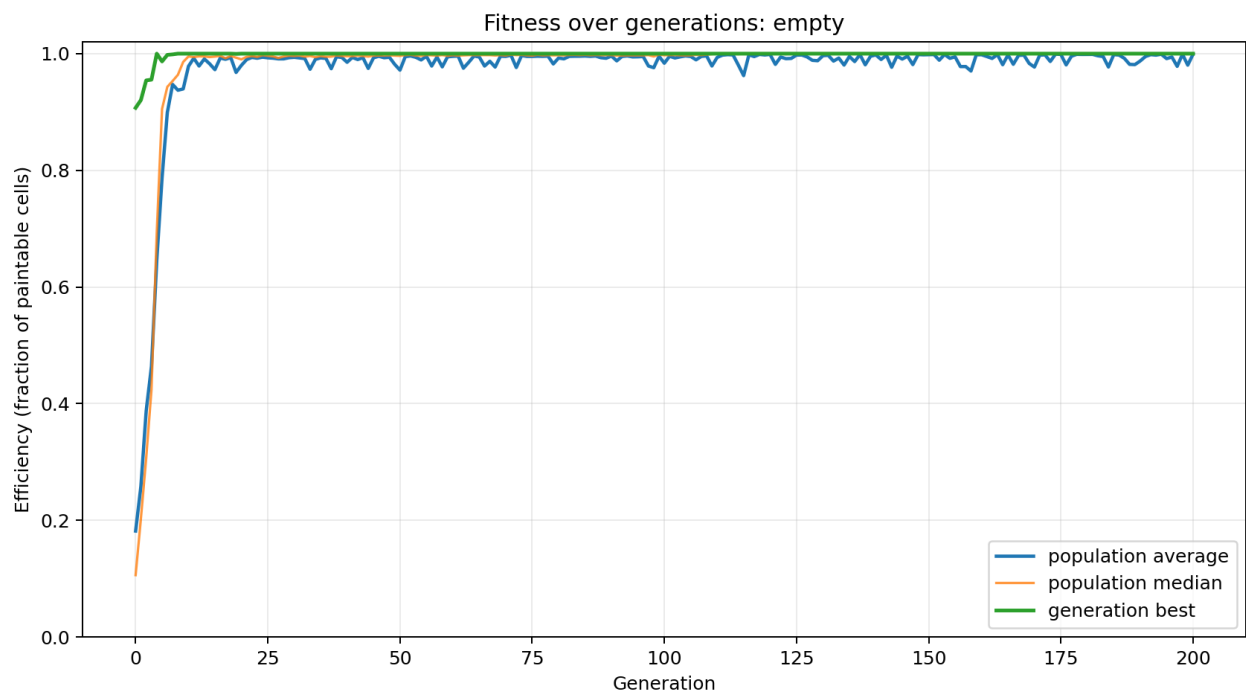


Figure 2: Empty-room fitness versus generation.

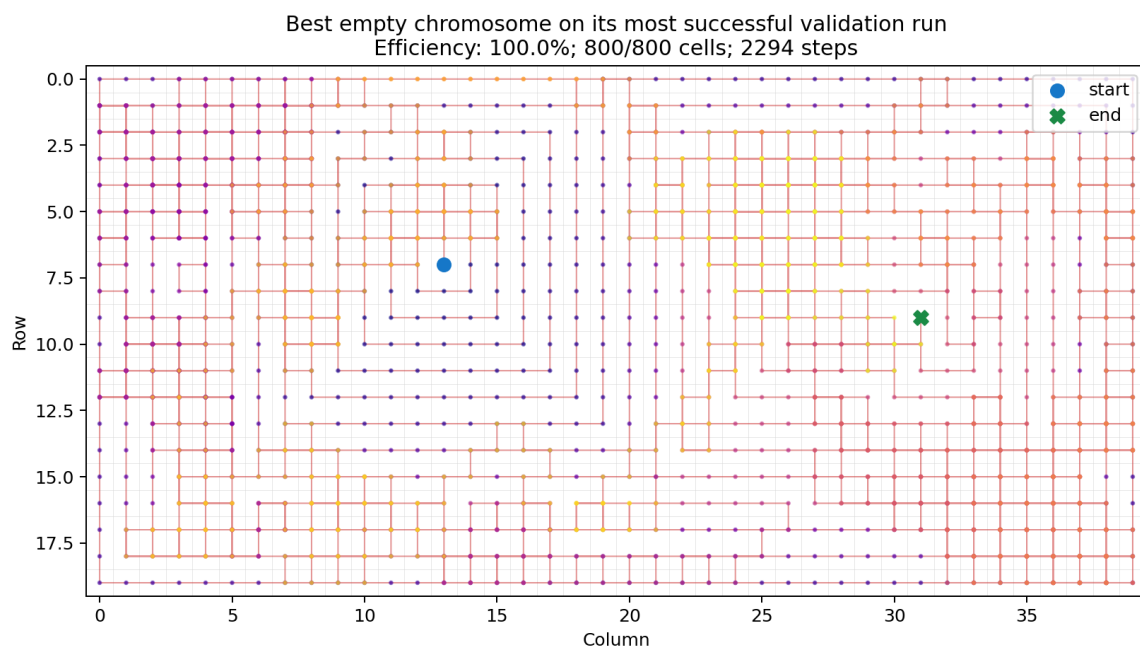


Figure 3: Trajectory of the empty-room champion on its most successful validation run.

6. Final chromosome population

The final population is shown as a heat map. Vertical bands indicate loci where selection has made the same action common across many chromosomes. Other loci remain more varied because several different local rules can produce similar coverage. The exact best chromosomes are stored in results/summary.json.

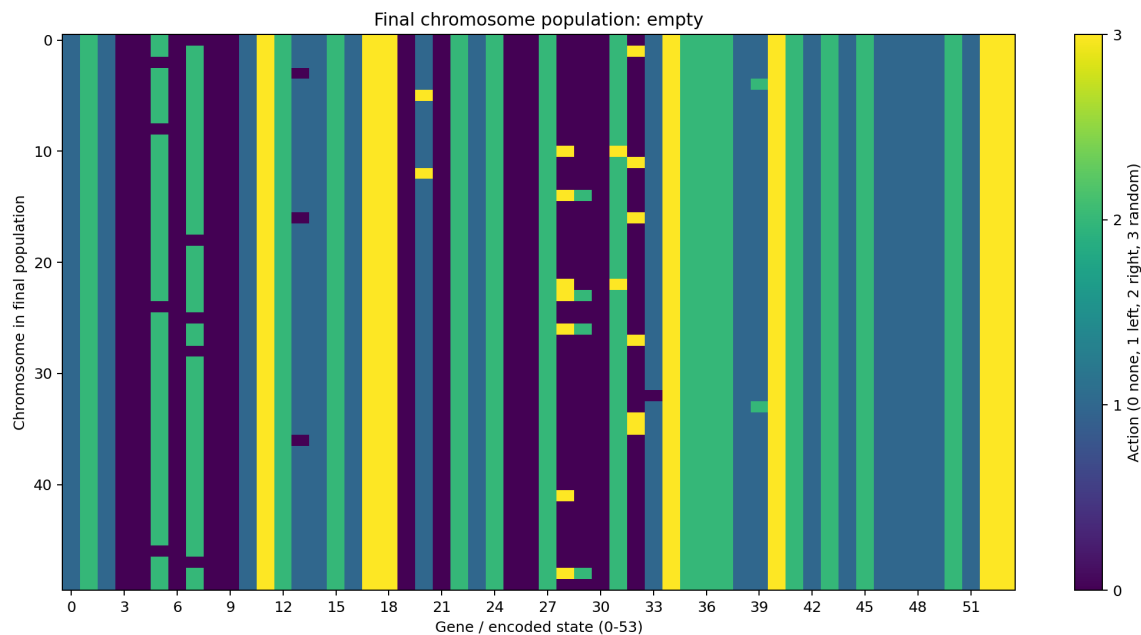


Figure 4: Final set of 50 chromosomes for the empty-room experiment.

7. Assignment 3: Strategic innovations

Two possible innovations that create a sudden fitness jump are:

- A rule that turns when the forward cell is blocked and then follows the open side of a wall, allowing the robot to continue instead of repeatedly pushing into the same obstruction.
- A pair of rules that uses painted neighbours as a local cue: it continues along an unpainted row, then turns at a painted boundary to enter the next unpainted strip. This approximates the serpentine sweep without global memory.

In the fitness plot, a jump occurs when a chromosome first combines enough of these local behaviours to stop wasting steps and reach previously missed regions. Because the chromosome is local and memoryless, the behaviour is not an exact planned route; it is an emergent pattern created by the interaction of painted-cell cues and walls.

8. Assignment 4: Furniture and transfer

The empty-room champion was first reused without modification in the furnished room. Furniture interrupts the long straight sweeps learned in the empty room. When the robot reaches an obstacle, a local turn rule can send it around the wrong side, repeat an already painted corridor, or leave a pocket behind the obstacle. This explains both the lower average efficiency and the large variation between start positions.

Experiment	Mean	Best	Worst	Paintable
Empty champion on furnished room	94.61%	99.57%	82.14%	700
Fresh GA on furnished room	98.89%	99.57%	97.29%	700

On the primary validation sample, transferring the empty-room chromosome gives a mean efficiency of 94.61%, while evolving from the beginning for the furnished room gives 98.89%. The fresh furnished-room policy is therefore about 4.29 percentage points better on that sample.

For a stricter paired comparison, both champions were evaluated on the same eight furnished-room starts. The transferred empty-room champion averaged 80.88%, including one run at 0.14%; the furnished-room champion averaged 98.89%. This confirms that the furniture-specific evolution is more robust, even though the exact score depends on the start position.

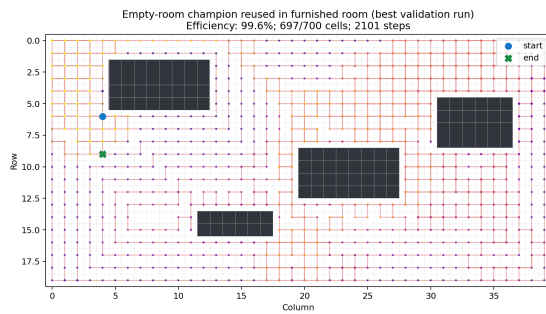


Figure 5: Successful transfer run.

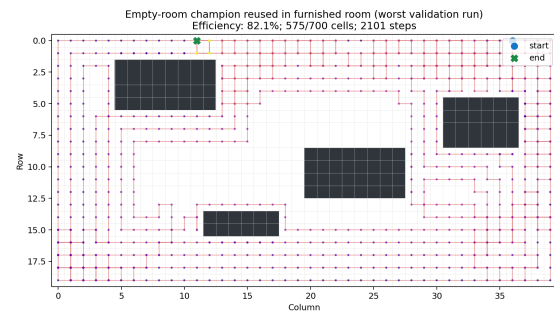


Figure 6: Lower-performing transfer run.

9. Furnished-room evolution from the beginning

When the furnished room is used during training from generation zero, selection rewards rules that respond to the furniture layout. The resulting curve rises quickly and then remains close to 0.99. The evolved trajectory shows repeated wall-aware detours around furniture, rather than assuming that every row is continuously traversable.

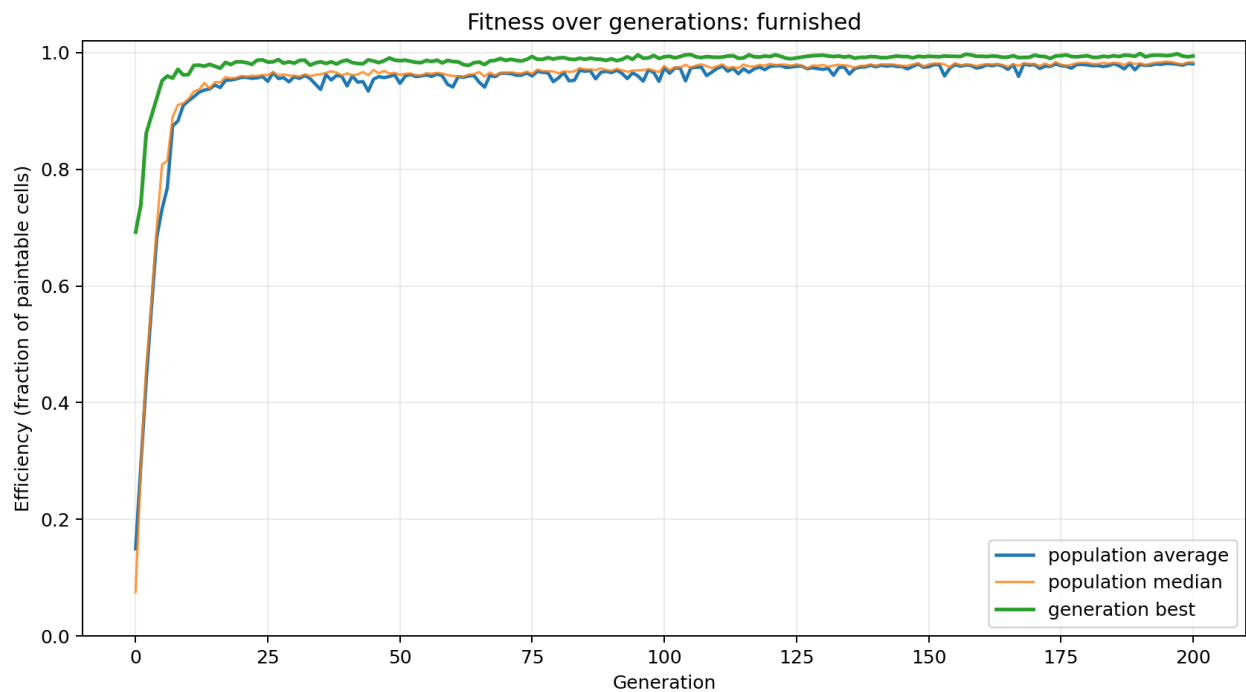


Figure 7: Fitness versus generation for the furnished-room GA.

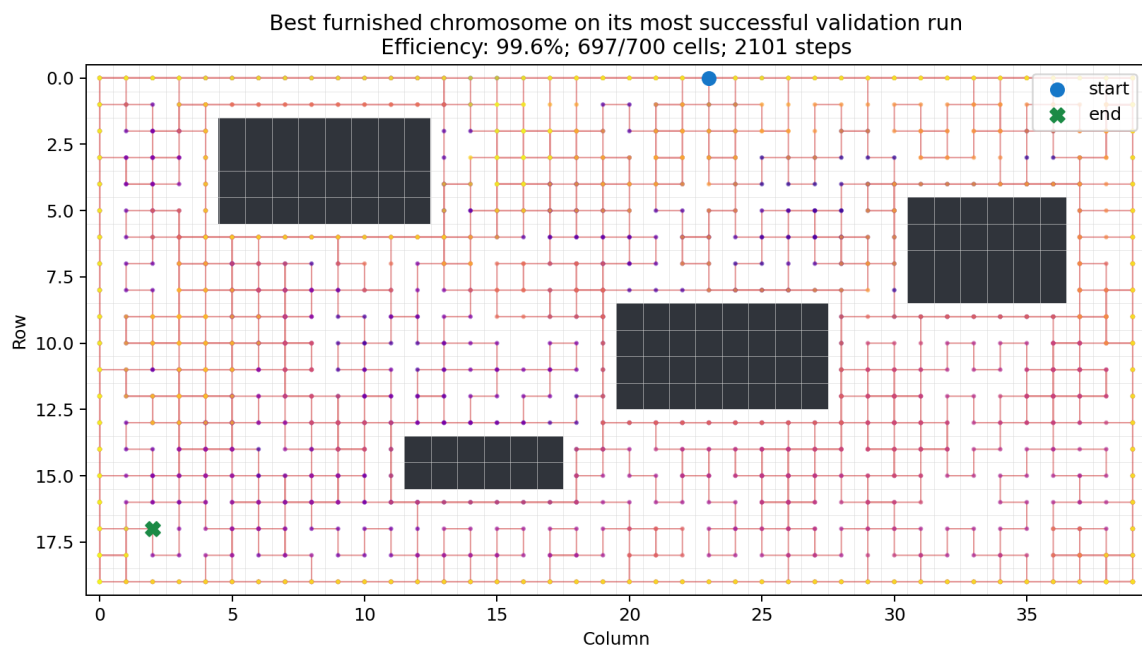


Figure 8: Trajectory of the best furnished-room chromosome on its most successful validation run.

10. Conclusion

The experiment demonstrates that a genetic algorithm can discover a useful local policy from only the four-cell state. In the empty room, the final champion covers nearly all floor space and can reach complete coverage on a validation start. However, the chromosome is specialized to the geometry it sees during evolution. Furniture changes the local state distribution and breaks the learned sweep, so direct transfer is less reliable. Re-evolving the same chromosome representation in the furnished environment recovers high efficiency. This supports the central genetic-algorithm idea: fitness-driven selection, crossover, and small mutations can discover environment-specific behaviour without manually writing the full strategy.

11. How to reproduce

```
python lab6_genetic_algorithm.py --output results
python -m py_compile lab6_genetic_algorithm.py
python make_lab6_report.py
```

The source code, figure files, CSV fitness histories, and JSON summary are included with this report. The PDF was rendered and inspected after generation for layout and legibility.